

Understanding Job Coaches' Perspectives on Using Virtual Reality as a Job Training Tool for Training People with Intellectual Disabilities

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Figure 1: VR Scenarios prototyped for the study: (a) Proper hand washing tutorial. The user clicks through the numbers to go through the tutorial. When a number is clicked, a video is displayed with the instructions. (b) Bathroom cleaning exercise. The user is required to clean the black stain on the counter using the proper tools. The bottom image shows the cloth on the stain when cleaning (c) User is asked a question by a customer and the responses are selected in the form of quizzes (the quiz is shown in the bottom image)

ABSTRACT

This study aimed to investigate the potential of virtual reality (VR) as a training tool for job coaches working with people with intellectual disabilities (ID). In this work, we conducted a qualitative study with job coaches of people with ID that included a workshop, design sprint, and interviews where existing and prototype VR apps aimed at job training people with ID were introduced and explored. The results of the study indicated positive feedback from the job coaches, with all stating that the VR had the potential as an effective training tool and that its ability to deliver customized experiences was appreciated. The study also highlighted the unique challenges and needs of trainees with ID and the importance of adapting training methods accordingly. Future research should aim to include a larger sample size, control group, long-term follow-up, more diverse

simulation scenarios, and study other disabilities and demographic groups. Overall, this study suggests that VR-based training has the potential as an effective tool for job coaches working with people with ID.

CCS CONCEPTS

• **Human-centered computing** → **Accessibility**.

KEYWORDS

virtual reality, job training, people with intellectual disabilities, accessibility

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1 INTRODUCTION

Virtual Reality (VR) has gained significant attention in recent years for its potential to provide immersive and interactive training experiences. In particular, VR has been suggested as a powerful tool for vocational training and employment of people with intellectual disabilities (ID) [6, 18, 19]. People with ID often face barriers to employment, such as a lack of appropriate job skills and difficulty with social interactions [4]. As such, VR has been explored as it offers a safe and controlled environment for individuals to practice and learn job skills, without the pressure of real-world consequences [6]. Previous research has also demonstrated that skills learned by people with ID in VR are effectively transferred to real-life settings [6, 10, 21].

With these motivations, recent research has started exploring job training with virtual reality for people with ID. Significantly, several studies have investigated VR-based social skills training [17, 20], which can be used for teaching individuals how to interact with coworkers and customers in a professional setting. In addition, a study by Smith et al. explored using VR to train people with ID on interview skills [26]. Furthermore, a more recent study investigated using VR to train six transferable skills that are useful in common jobs [6].

While much of such previous research has worked directly with people with ID as their end user, there is a limited focus on understanding the experience or the expectations of the job coaches who train people with ID on introducing such new technologies to the job training process. Therefore, this paper focuses on understanding the effectiveness of virtual reality in training people with ID from the perspective of job coaches. This exploratory study addresses the research question: "What are the experiences and expectations of job coaches on using VR for the job training process of people with ID, and how can VR interventions help improve this experience?". As such, our main objective was to gain job coaches' insights on using VR as a potential technology for the job training process of people with ID. To address this research objective and questions, we conducted a study that included a focus group, a design sprint, and an individual interview with five job coaches who train people with ID for employment. As our contribution, we present the findings from our study that detail the experiences of the job coaches using the VR prototype scenarios, their expectations, goals, etc., and some example scenarios proposed during the study.

2 METHODOLOGY

The study aimed to understand different experiences and factors influencing building accessible scenarios in virtual reality from a job coach's perspective. The study was conducted in three phases (1) A contextual inquiry with a focus group study that introduced existing and prototype scenarios of job training methods in virtual reality (2) A design sprint to gather information about potential future expectations and scenarios (3) interviews of job coaches to understand the behavioral and cognitive aspects of the training process of people with ID.

2.1 Participant demographics and recruiting

The study was conducted with five participants (Age: 25-67; 1 Female, 4 Male) recruited from a local organization that trains people

with ID for employment. The participants included five job coaches, and one was the organization's director, all of them provide employment training to people with ID. Participants had a wide range of training experiences from as few as two to more than ten years. One out of five had previous VR experience, while three were novices. The participants were recruited through a survey that was sent to the organization which also collected their demographic information. The participants were given informed consent before the beginning of the study.

2.2 Apparatus

The study sessions were conducted utilizing an Oculus Quest 2 VR headset and were audio and video recorded. To enhance the study, pre-existing VR apps, including Job Simulator¹ was installed on the headset. Additionally, in collaboration with the director of the organization, three bespoke VR training scenarios were developed using GMetri and Spatial.io software to showcase the capabilities of VR and elicit feedback from participants during the Phase 2.

Scenario 1: This scenario (Fig. 1(a)) included a tutorial component in which participants were presented with a step-by-step visual guide on properly washing hands using virtual reality (VR). Following the tutorial, participants were then directed to a subsequent scenario in which they were expected to apply the knowledge they acquired from the tutorial. The hand-washing task consisted of four steps: wetting hands, applying soap, rubbing hands together, rinsing hands, and drying hands with a towel. Participants' performance in this task was evaluated using a scoring system, with points awarded based on their adherence to proper hand-washing techniques.

Scenario 2: This scenario (Fig. 1(b)) included a task where participants were instructed to perform a bathroom cleaning exercise. The task involved switching on a faucet, obtaining a cleaning cloth, applying soap, and utilizing the cloth to clean stains present in the bathroom. The objective was to evaluate the participants' ability to clean a bathroom effectively.

Scenario 3: This scenario (Fig. 1(c)) aimed to investigate the responses of retail store employees in simulated customer interactions. Participants were presented with a series of situations and instructed to respond in the form of quizzes about how they would react in those situations. The scenarios included conversations with customers, peers, and managers/employers. The objective was to assess the participants' responses in these simulated situations.

The study employed three simulation scenarios to target different aspects of training. The first scenario evaluated the participants' understanding and interpretation of instructions. The second simulation assessed the participant's ability to apply their acquired knowledge. This simulation evaluated the participants' technical skills and tested gesture control. The third simulation evaluated behavioral training. The overall objective of these simulations was to expose participants to various training methods and assess the impact of the acquired training in real-time situations.

¹<https://www.oculus.com/experiences/quest/3235570703151406/>

2.3 Procedures

2.3.1 Phase 1: Virtual reality interaction workshop with the focus group. The workshop aimed to expose job coaches to VR and understand their interactions with this technology and their perspective on whether this tool effectively trains people with ID. The experience began with them completing the initial environment settings. In the first part of the workshop, the participants were allowed to interact with an existing virtual reality environment- the Oculus Job Simulator for 5 minutes. Then the participants were allowed to experience the three demo training scenarios afterward. While in the environments, they were asked to think aloud about what they were seeing, thinking, and doing. They were asked questions about their decisions while doing the tasks, what enhanced their experiences, and what they found challenging. After completing four apps/scenarios, the participants were asked about their experience in those simulations, what part of the training they liked the most, what felt easy, and what was difficult to understand or perform.

2.3.2 Phase 2: Design sprint. After completing Phase 1, participants were invited to participate in a design sprint divided into three steps that lasted 30 minutes. The design sprint focused on strategizing the building of the VR simulator for training people with ID through understanding various user needs and project logistics. The participants were provided with blank sheets of paper, each divided into three sections labeled "long-term goals," "challenges," and "potential scenario ideas." In the first step, participants were asked to jot down their long-term plans for the study. They were prompted to consider what they envisioned as the outcome, what future work would be required, and what solutions they hoped to achieve.

In the next step, participants were asked to identify the potential issues and challenges that could arise while developing the virtual reality simulation for the intellectually disabled. This included the challenges they currently face while training and what challenges they anticipate would arise while using this technology.

Finally, in the last step, participants were asked to write down potential scenarios that could be built for training. They were also asked to note down the areas/ domains in which the training needs to be given the most. They briefly noted down the training modules they felt that could be designed for producing the simulations. Each of these three tasks was given 5 minutes to complete, with an additional 5-minute break in between each task.

2.3.3 Phase 3: Interviewing. After completing the initial two phases of the workshop, the study continued with individual interviews with the job coaches. These interviews were conducted during the following week over four days, lasting 60 minutes each. These interviews aimed to gain behavioral insights into the coaches' overall training backgrounds, experiences, and current training methodologies and modules. The interviews were conducted via Zoom, and participants were provided with informed consent forms before the start of the interview.

3 FINDINGS

The data from all phases were collected, transcribed and analyzed together to identify themes using the Braun and Clark's thematic

mapping approach [8]. Three main themes were identified (Figure 2): Job training process, Feedback on VR Scenarios, and Insights for new scenarios.

3.1 Job training process

The majority of the data under this theme was gained through the interviews of the participants, where they described their training processes, how they handle them, and the characteristics of the individuals they train.

Coaches' strategies with trainees: All participants agreed that every person with ID is unique, with different challenges and needs, and that the coaches must adapt their training methods accordingly. The main tasks of the coaches are to help trainees acquire life skills such as cleaning their apartments, shopping, using computers, applying for jobs, and using public transportation. *"Some people need my assistance in computers, showing them how to apply for jobs online."* (P2) All the job coaches added that they loved working in the organization as they had the urge to help. *"I always wanted to help people with disabilities because I always had a soft spot for them, so I chose this organization."* (P1)

Training process: The training process primarily consists of one-on-one sessions, with the potential for expansion to small group sessions of up to eight individuals. Training sessions range from 30 minutes to one hour, with the frequency and duration being contingent upon the trainee's progress and response. The primary objective of the coaches is to facilitate acclimation to the job and promote cohesion with coworkers and managers while also providing education on the management of individuals with disabilities. *"I get them to learn how to meet their new co-workers, build their relationship with them and their supervisors."* (P2) Observations are conducted to assess progress, and support is reduced subsequently on successful improvement in performance after training. Each trainee receives individualized training tailored to their unique challenges and needs, including accommodations such as audio/visual cues and alternative instruction methods. The people they support exhibit a range of abilities, including proficiency with technology or limitations in motor control. One of the coaches added, *"For employees with intellectual disabilities, everything is black and white. They don't understand the grays of the job spaces they work in"* (P3) suggesting the need for simplified instructions for actions. Some of the trainees also lack access to technology due to their disabilities. They often face societal rejection and hence lack self-confidence. In instances of bullying, coaches intervene to manage the situation. *"I educate the business on how to manage maybe somebody who you know has a disability."* (P5)

3.2 Feedback on VR scenarios:

The participants gave feedback about all the virtually simulated experiences that they experienced. This feedback was collected during the workshop and through the interviews conducted later.

General feedback: Only one participant was an experienced virtual reality gamer. Despite the lack of prior experience, all participants could quickly complete the initial setup. They found the experience to be exciting and believed it had the potential to be used as an effective training tool. The participants appreciated the ability to deliver customized experiences.

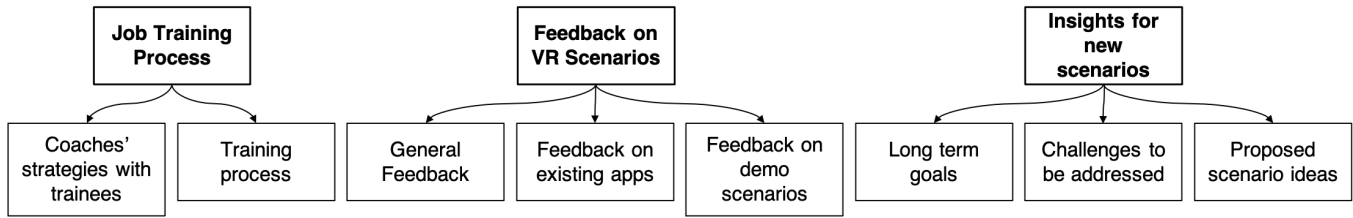


Figure 2: Thematic map of the findings.

Feedback on existing VR applications: The scenarios began with the currently available "Job Simulator" app by Oculus. The overall feedback from the participant about this app was that the tasks presented were overwhelming. On average, participants completed three to four tasks during their 5-minute exposure to the simulation. While the tasks were overwhelming, the participants appreciated the social interactions presented in the app. P5 said, "The unhappy manager, taking coffee breaks, and throwing paper chits at their colleagues made the office feel real and should be included in the scenarios."

Feedback on demo scenarios: After experiencing the "Job Simulator" simulations, participants were presented with three scenarios, as described in Section 2.2. In Scenario 1 for washing hands, participants were asked to perform the activity by selecting the proper order of events as a quiz. All participants were able to point and choose the proper objects correctly. All the participants felt that the step-by-step tutorials would benefit the individuals they support and encouraged the development of similar training modules. The participants also appreciated the quiz-based evaluations where P2 said "The quizzes really helped directly apply what I learned in the tutorial. I think such activities would help in understanding the content better." However, 2 out of 5 participants reported feeling uneasy about looking down at the floor, as they felt it would affect their balance and suggesting to add representation of the body in the VR scenarios, such as hands and legs. In the third scenario, the participants were exposed to various situations at a retail store, where they were supposed to interact with peers and angry customers. The participants stressed the need for such training and evaluations based on users' reactions in such situations.

3.3 Insights for new scenarios

This theme describes the insights and expectations that were discussed for building new training scenarios and simulations from the job coaches' perspective.

Long-term goals: Several expectations were discussed as long term goals. The simulation could be utilized for evaluation purposes, including assessing skills and comprehension. It also could be designed to be engaging and enjoyable, encourage regular use, and closely mimic real-life job environments to provide users with a realistic experience. The simulation should be more interactive for interactive in-job activities. "The strategic goal is to have all the job coaches trained on the technology." (P5) Future goals include the development of a library of jobs and tasks that align with users' interests and the potential inclusion of social situations.

Challenges to be addressed: The participants discussed challenges faced by job coaches when working with people with ID that should be addressed in future scenarios. These include keeping the user engaged in the training process due to minimal attention spans, navigating the unique abilities and disabilities of each individual, and physical challenges such as mobility and coordination, motor control, spatial balance, and sensory issues that can make training difficult. As P3 said, "They are predominantly visual learners, and hence it becomes difficult for explaining things verbally", suggesting the need for reducing distractions in simulations so that it helps in improving their focus.

Proposed scenario ideas: Proposed scenario ideas were further classified as general scenario activities that targeted job-specific activities, life skills scenarios focusing on training them in life skills needed for employment, and behavioral scenarios to train them in behavioral aspects at job spaces.

- **General scenario activities:** To effectively train people with ID, job scenarios can be built based on the location of specific jobs, such as retail stores, restaurants, and offices. These scenarios can include various activities relevant to the job, such as sorting, stocking, handling money, cleaning, and safety. For example, a retail store scenario could include stocking shelves, handling money, and cleaning, while a restaurant scenario could include washing dishes, food prepping, and food safety. Similarly, a bakery scenario could include folding boxes, putting labels on containers, etc. Cleaning activities, such as vacuuming, laundry, emptying trash bins, and dusting, can also be included in these scenarios. Additionally, interactive elements such as "Wrong vs. right" choices and "choose your adventure" activities can be incorporated to keep participants engaged. According to feedback from participants, dishwashing activities were particularly popular.
- **Life skills scenarios:** In addition to job-specific skills, people with ID must develop specific life skills to succeed in the workforce. These life skills include being able to get ready and arrive on time, choosing appropriate clothing to wear, communicating effectively about sick leaves and other time off, and understanding how to clock in and out correctly. These skills are essential for employees in any job and will help them to function independently and responsibly in the workplace. Moreover, VR gives the advantage of training certain things that can become difficult in otherwise real situations. "VR simulators could help us have difficult/ uncomfortable conversations like hygiene." (P3) This opens another channel for designing scenarios.

- Behavioral scenarios: In addition to job-specific and life skills, it is also important to include activities that promote or discourage specific behavioral responses to reinforce professional behaviors in people with ID. This includes addressing and discouraging behaviors such as throwing things at co-workers or playing computer games during work hours by incorporating prompts such as being scolded by bosses or losing hours. Situational scenarios should also be included, such as dealing with upset customers, taking orders from bosses or clients, handling workspace conflicts, managing work-related stress, and appropriately responding to requests for help. One of the participants mentioned the need to create scenarios that train people with ID to understand the complexity of situations and correctly respond to them. *"If working and a coworker interrupts, have text/audio options how to respond. So that you can stay focused on your tasks."* (P5) These activities will help to prepare people with ID for real-world work scenarios and reinforce appropriate professional behaviors.

4 DISCUSSION

In this qualitative study, we investigated the potential of VR as a training tool for job coaches working with people with ID. The VR scenarios received generally positive feedback from participants, with all stating that the study had the potential as an effective training tool and that the ability to deliver customized experiences was appreciated. Next, we discuss some of the findings.

Participants found the "Job Simulator" app overwhelming and suggested that a simpler module with direct instructions would be more appropriate for the target population. This indicates that while designing simulations, the cognitive load that instructions can produce and the intellectual challenges that people with ID face, such as reduced understanding of multi-task instructions or indirect instructions, should be assessed. Sound design methodologies focusing on improving usability and accessibility would lead to building better simulating experiences that all types of people with ID could use. [1, 14]. Participants also reported feeling uneasy about looking down at the floor, as it affected their spatial balance. On the other hand, the step-by-step audio/video tutorial in VR for handwashing received positive feedback, with participants stating that it would benefit the individuals they support and encouraged the development of similar training modules. The washing of hands tutorial, followed by its assessment through pointing and touching objects in the correct order, generates a sense of involvement stimulated through the "virtual touch" of things [3, 30]. To address the shortcomings of current simulations, we suggest that the content of future VR training modules for people with ID should be simple, with a reduced cognitive load and direct call to action, to prevent users from getting overwhelmed [24, 25]. The modules should also be short, with constant feedback through pop-ups, scores, suggestions, and alerts with enhanced substitute cues to keep the users' attention intact [9, 12, 22, 30].

Our study also highlighted the unique challenges and needs of trainees with intellectual disabilities and the importance of adapting training methods accordingly. Trainees exhibit a range of attributes, including proficiency with technology and limitations in

motor control, while some lack access to technology due to their disabilities. Additionally, due to the physical challenges faced by people with ID, VR simulations should be designed considering their motor control, spatial balance, sensory, mobility, and coordination issues [11, 13, 27]. This could be achieved by studying the impact of gesture-control, and click-based or non-gesture-control interactions [7, 15].

Furthermore, the study highlighted the need for training in life skills and behavioral aspects, such as self-regulation and temper control. They also face societal rejection and a lack of self-confidence. Developing conversation-based training modules can help enhance communication skills and self-confidence working on their behavioral regulations [2, 5, 23]. Users can practice responding to situations in a realistic environment by providing conversational training through virtual avatars. The use of virtual avatars can be made more realistic by changing the tone and language of the avatars. Virtual reality training has shown significant improvement in community-based activities and skills. Using speech recognition, virtual conversations with naturalistic feedback can help candidates practice their interview skills [16, 28, 29]. Training through role-plays in virtual reality environments provides a safe space for practicing real-life scenarios with varying degrees of emotional impact, such as dealing with angry or frustrated customers or responding to disturbing co-workers. [16, 28]

The data were collected separately from all the stages of the study and analyzed thematically together, producing some design insights that would define the further building of the simulator. As our recommendations, considering the challenges of the target users, the simulator should be designed as simple and intuitive with direct call-to-action features with minimal multi-step task procedures. As many participants stressed the range of allied challenges they come across in their trainees, it is necessary to understand that the uniqueness of users and the platform should be made inclusive for handling these challenges. Moreover, although the simulator focuses on training trainees with job skills, there are certain life skills that employees require to learn, which can enhance their productivity before and after their job hours. Identifying these activities would enhance their self-confidence in the job spaces. Increasing users' interactivity in the environments through activities that enhance the virtual touch would make the training more fun and efficient. These interactions would feel more real by including the real elements in the virtual spacing, enhancing the realization of simulations.

5 LIMITATIONS AND FUTURE WORKS

The study had some limitations that should be considered when interpreting the findings. The sample size was small, with only 5 participants from one local organization, limiting the findings' generalizability. Our future studies aim to include a larger sample with more diverse participants such as participants from different organizations and also participant train people with ID for different skills, etc.

The participants also had limited exposure to virtual reality, which may have affected their ability to engage with the simulations and provide accurate feedback. This also brings to question the long terms effects of VR interventions. As such, our future research

directions will conduct follow-up assessments to evaluate the long-term impact of virtual reality training.

The scenarios that were prototyped for the study were preliminary due to the study's exploratory nature. Therefore, we aim to use the findings from our study to improve the developed scenarios and create new scenarios to explore with the involvement of the job coaches and their trainees. We suggest that future VR training modules for people with ID should be designed with simplicity, with a reduced cognitive load and direct call to action to prevent users from getting overwhelmed. The activities should also be short, with feedback in the form of pop-ups, scores, suggestions, and alerts to keep the users' attention intact. The simulations should also consider the unique challenges and needs of trainees with intellectual disabilities, such as limitations in motor control and lack of access to technology. In addition to addressing technical challenges, the VR simulations should also focus on training in life skills and behavioral aspects, such as self-regulation and temper control. Developing conversation-based training modules using avatars can also help enhance communication skills and self-confidence. The study also highlights the importance of communication skills in improving the self-confidence of people with ID. Future work should also include investigations into the impact of gesture-control interactions vs. click-based or non-gesture-control interactions and the design of VR simulations that provide users with more ways to interact with other users. This could include using display trackers, speech recognition, gesture recognition, avatars, and auto-navigations.

6 CONCLUSION

This study examines the potential of virtual reality (VR) as a training tool for job coaches working with people with ID. For this, we conducted a study that included a focus group workshop, a design sprint, and individual interviews with five job coaches who train people with ID recruited from a local organization. This study's findings revealed insights into the nature of current training practices through the job coaches who train people with ID. The findings also highlight the potential of virtual reality as a training tool for job coaches working with people with ID and suggest areas for future research and development. With the right design and approach, VR simulations can provide an engaging and effective way to train and support people with ID in their work and life skills.

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